MAGNESIUM SULFATE

High-alert drug

Antidote: calcium gluconate

CHEMICAL NAME $MgSO_4 \cdot 7H_2O$ (magnesium sulfate heptahydrate)
CLASS Electrolyte; anticonvulsant in eclampsia; neuroprotectant
TRADE NAMES Magnesium sulfate 50% injection · Epsom salt (oral) · $MgSO_4$ 4 g/100 mL premix

THERAPEUTIC USES / INDICATIONS

- Prevention and treatment of eclamptic seizures (severe pre-eclampsia/eclampsia).
- Fetal neuroprotection in preterm birth <30–32 weeks.
- Acute severe asthma (single IV dose).
- Hypomagnesaemia; torsades de pointes; arrhythmias with digoxin toxicity.

ADMINISTRATION

- **IV** loading dose over 5–15 min, then maintenance infusion via pump; dilute 50% solution (max 20% for IV).
- **IM** (Pritchard regimen): deep, split into each buttock, with lidocaine.
- Use pre-mixed bags where available; double-check concentration.
- Keep calcium gluconate 10% 10 mL at bedside.

DOSAGES

- **Pre-eclampsia/eclampsia (Zuspan):** 4 g IV over 5–15 min, then 1 g/h for 24 h after delivery or last seizure; recurrent seizure: further 2–4 g over 5 min.
- **Pritchard IM:** 4 g IV + 10 g IM (5 g each buttock), then 5 g IM every 4 h.
- **Asthma:** 1.2–2 g IV over 20 min (child 40 mg/kg, max 2 g).
- **Torsades:** 2 g IV over 10–15 min (faster in cardiac arrest).
- **Hypomagnesaemia:** slow IV infusion; up to 40 g (160 mmol) over up to 5 days may be needed (reduce in renal impairment).

PHARMACOKINETICS

- **Onset:** IV immediate; IM 1 h. **Duration:** IV ~30 min; IM 3–4 h.
- Not metabolised.
- Excreted by kidneys — accumulates in renal impairment/oliguria.
- Effective anticonvulsant level 2.5–7.5 mEq/L (≈1.25–3.75 mmol/L, US label); monitor reflexes, RR and urine output.

PHARMACODYNAMICS

- Blocks **NMDA receptors** and **calcium channels** → CNS depression, raised seizure threshold; cerebral vasodilation.
- ↓ ACh release at neuromuscular junction → muscle weakness.
- Bronchial smooth-muscle relaxation; membrane stabilisation in heart.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Flushing, warmth, nausea, headache.
- Muscle weakness, drowsiness.
- Injection pain (IM).

ADVERSE EFFECTS (SERIOUS)

- Magnesium toxicity: reflexes fall above 4 mEq/L (2 mmol/L) and are lost near 10 mEq/L (5 mmol/L) with respiratory paralysis; >12 mEq/L (6 mmol/L) may be fatal.
- Hypotension, bradycardia, heart block.
- Hypocalcaemia; neonatal hypotonia/respiratory depression.

CONTRAINDICATIONS & CAUTIONS

- Myasthenia gravis.
- Heart block (except torsades).
- **Caution:** renal impairment (reduce maintenance), oliguria.

KEY INTERACTIONS

- **Nifedipine, calcium-channel blockers:** profound hypotension, neuromuscular block.
- **Neuromuscular blockers:** prolonged paralysis.
- **Aminoglycosides:** ↑ neuromuscular weakness.

NURSING MANAGEMENT

- Hourly: deep tendon (patellar) reflexes, RR (>12/min), SpO_2, urine output (>30 mL/h or >100 mL/4 h), BP, level of consciousness.
- Stop infusion and call doctor if reflexes absent, RR <12, urine output low.
- Toxicity antidote: calcium gluconate 10% 10 mL IV over 10 min.
- Measure Mg^{2+} levels if renal impairment or toxicity suspected.
- Continuous CTG; inform neonatal team.

PATIENT EDUCATION

- You may feel warm and flushed when the drip starts; this is expected.
- Report breathing difficulty, heavy weakness or drowsiness.

KEY POINTS

- Drug of choice for eclampsia: 4 g load, 1 g/h × 24 h.
- Toxicity: reflexes lost first → respiratory arrest.
- Hourly reflexes, RR, urine output.
- Antidote: calcium gluconate 10% 10 mL.

SOURCES CHECKED **[673]** Magnesium Sulfate 10% w/v solution for infusion (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158) **[311]** NICE NG133 **[674]** Magnesium Sulfate, IM, IV (US label)
Full references p. 432–457